SIMT Architecture

Thursday, January 29, 2026 4:28 PM

ThreadIdx.x

```
/*0000*/  /*0x1100e804      */   MOV32 R1, g [0x4];                /*base */
/*0004*/  /*0x1100ea08      */   MOV32 R2, g [0x5];                /*base */
/*0008*/  /*0x6004000500000003*/ IMAD32I.U16 R1, R0L, 0x4, R1;     /*index */
/*0010*/  /*0x6004000900000003*/ IMAD32I.U16 R2, R0L, 0x4, R2;     /*index */
/*0018*/  /*0xd00e020d80c00780*/ GLD.U32 R3, global14 [R1];         /*load*/
/*0020*/  /*0xd00e040980c00780*/ GLD.U32 R2, global14 [R2];         /*load*/
/*0028*/  /*0x1100ec04      */   MOV32 R1, g [0x6];                /* base */
/*002c*/  /*0xb0020608      */   FADD32 R2, R3, R2;                / * add */
/*0030*/  /*0x6004000500000003*/ IMAD32I.U16 R1, R0L, 0x4, R1;     /* index */
/*0038*/  /*0xd00e0209a0c00781*/ GST.U32 global14 [R1], R2;         /* store */
```

R1 = R0 * 4 + R1

(8x1) → (8, 1)

Warp size = 4, TB size : 8, 2 warps. vector size : 8

Architectural state. (mem. RF. PC. special purpose Register)

C[tid] = A[tid] + B[tid]

↙ x3000 ↓ x1000 ↓ x2000

ARF

warp size

W0

R0	0	1	2	3
1	x1000	x 1000→1004	x 1008	x 100C
2	x 2000	x 2004	x 2008	x 200C
3				

PC = x18

W1

0	4	5	6	7
1	x1000	x 1000	x 1000	x 1000
2				
3				

PC = 4

TB (8, 1)

TB (4, 2)

ThreadIdx.x = 0 1 2 3

ThreadIdx.y = 0 0 0 0 / 1 1 1 1

TB (2, 4)

TB (1, 8)

PRF

TB: W0 ARF, W1 ARF

TB1

